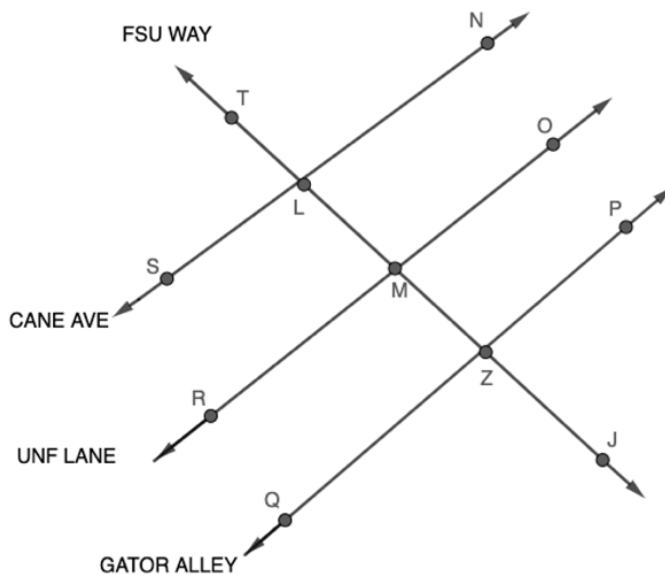


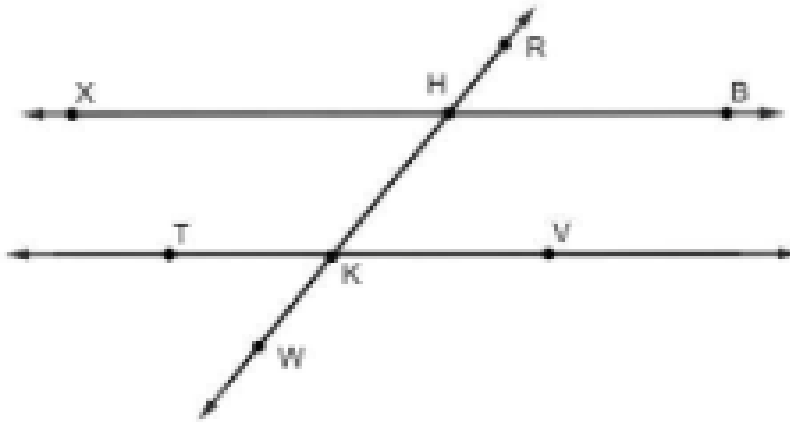
Use the following information for questions 1-3. FSU Way, a transversal, TJ , crosses the streets Cane Avenue, UNF Lane, and Gator Alley.



Determine the value of the variables and each angle measure given Cane Avenue, UNF Lane, and Gator Alley are parallel to each other.

1. $\angle TLN = (3x - 18)^\circ$, $\angle MZQ = (5x + 14)^\circ$, and $\angle NLM = y^\circ$
 $3x - 18 + 5x + 14 = 180$
 $8x - 4 = 180$
 $8x = 184$
 $x = 23$
 $3(23) - 18 = 69 - 18 = 51^\circ = m\angle TLN$
 $5(23) + 14 = 115 + 14$
 $= 129^\circ = m\angle MZQ$
 $180 - 51 = 129 = y$
 $129^\circ = m\angle NLM$
2. $\angle JZQ = (9x + 9)^\circ$ and $\angle TLN = (8x + 19)^\circ$
 $9x + 9 = 8x + 19$
 $x = 10$
 $9(10) + 9 = 90 + 9 = 99^\circ = m\angle JZQ$
 $8(10) + 19 = 80 + 19 = 99^\circ = m\angle TLN$
3. $\angle ZMR = (x + 14)^\circ$, $\angle LMO = (2x - 6)^\circ$, and $\angle LMR = y^\circ$
 $x + 14 = 2x - 6$
 $20 = x$
 $20 + 14 = 34^\circ = m\angle ZMR$
 $2(20) - 6 = 40 - 6 = 34^\circ = m\angle LMO$
 $34 + y = 180$
 $y = 146$
 $m\angle LMR = 146^\circ$

Use the following information for questions 4-7. Given $\overleftrightarrow{XB}$, $\overleftrightarrow{TV}$, and $\overleftrightarrow{RW}$ are shown. $\overleftrightarrow{XB} \parallel \overleftrightarrow{TV}$ where $\overleftrightarrow{RW}$ is a transversal.



Determine the value of the variables and each angle measure.

4. $m\angle TKW = (8x - 20)^\circ$ and $m\angle XHK = (3x + 2)^\circ$

$$8x - 20 = 3x + 2$$

$$5x = 22$$

$$x = 4\frac{2}{5}$$

$$8\left(\frac{22}{5}\right) - 20 = 35\frac{1}{5} - 20 = 15\frac{1}{5}^\circ = m\angle TKW$$

$$3\left(\frac{22}{5}\right) + 2 = 13\frac{1}{5} + 2 = 15\frac{1}{5}^\circ = m\angle XHK$$

5. $m\angle RHB = (2x - 20)^\circ$ and $m\angle XHK = 44^\circ$

$$2x - 20 = 44$$

$$2x = 64$$

$$x = 32$$

$$m\angle RHB = 44^\circ$$

6. $m\angle HKT = (3y - 18)^\circ$, $m\angle VKW = (8x + 12)^\circ$, and $m\angle RHB = (2x + 18)^\circ$

$$8x + 12 + 2x + 18 = 180$$

$$10x + 30 = 180$$

$$10x = 150$$

$$x = 15$$

$$2(15) + 18 = 30 + 18 = 48 =$$

$$m\angle RHB = 48^\circ$$

$$3y - 18 = 8x + 12$$

$$3y - 18 = 8(15) + 12$$

$$3y - 18 = 120 + 12$$

$$3y - 18 = 132$$

$$3y = 150$$

$$y = 50$$

$$m\angle VKW = 132^\circ$$

$$3(50) - 18 = 150 - 18 = 132^\circ = m\angle HKT$$

7. $m\angle BHK = (8x + 12)^\circ$ and $m\angle VKW = 140^\circ$

$$8x + 12 = 140$$

$$8x = 128$$

$$x = 16$$

$$8(16) + 12 = 128 + 12 = 140^\circ = m\angle BHK$$